

Simulation Data Transformation Module (SDTM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Subcategory: Simulation Governance

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Governed Space: Simulation Data Transformation

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Canonical Language: English (UCL™)

Governed Space

Simulation Data Transformation

Canonical Definition

Simulation Data Transformation Module (SDTM) defines the governance framework through which identified Simulation Inputs may be cleaned, filtered, normalized, converted, aggregated, disaggregated, interpolated, extrapolated, encoded, imputed, derived, synthesized, resampled, aligned, or otherwise transformed before or during simulation use while preserving the traceability, meaning, limitations, uncertainty, provenance, and material consequences of those transformations.

SDTM establishes the transformation layer of SIDS by governing the transition:

Source Information → Transformation → Simulation-Usable Input

so that transformed information cannot silently acquire properties, precision, representativeness, certainty, or observational status that the transformation process did not legitimately establish.

Operational Role

SDTM serves as the third internal governance space of the Simulation Input & Data Standard (SIDS).

The preceding modules establish:

SIIM — What information is entering the simulation?

SDRM — What reality can that information legitimately represent?

SDTM asks:

What has been done to that information before the simulation consumes it, why was the transformation necessary, and what did the transformation materially change?

The governed progression is:

Identified Input → Transformation Need → Transformation Method → Transformation Execution → Transformed Input → Transformation Effect Assessment → Transformation Status → Simulation-Usable Input

Module Operational Space

SDTM governs: transformation identification, transformation purpose, transformation necessity, transformation methods, transformation rules, transformation parameters, data cleaning, filtering, normalization, standardization, unit conversion, format conversion, aggregation, disaggregation, interpolation, extrapolation, resampling, encoding, decoding, feature derivation, imputation, synthetic supplementation, dimensional transformation, temporal alignment transformation, spatial transformation, coordinate transformation, data fusion, data reduction, outlier treatment, missing-value treatment, transformation sequencing, transformation dependencies, transformation reproducibility, transformation reversibility where relevant, information loss, transformation-induced bias, transformation-induced uncertainty, transformation limitations, transformation validation references, transformation versioning, transformation change, transformed-input identity, and transformation traceability.

Module Function

SDTM exists to prevent: transformed information being treated as raw information, cleaning decisions disappearing from the simulation record, filtering silently removing material observations, normalization changing meaning without disclosure, unit conversion errors, aggregation hiding material variation, interpolation being represented as observation, extrapolation being represented as measured reality, imputed values appearing as recorded values, synthetic supplementation appearing as empirical data, transformations introducing hidden bias, precision being artificially increased, information loss becoming invisible, multiple transformation stages becoming unreconstructable, changed transformation logic being applied under an unchanged identity, and simulation outputs becoming impossible to trace back to the information basis from which they were produced.

Simulation Data Transformation

A Simulation Data Transformation is any governed operation that materially changes the: representation, structure, scale, format, resolution, composition, value, distribution, temporal organization, spatial organization, completeness, or interpretation

of information before or during simulation use.

Conceptually:

Input A → Transformation T → Input B

Input B may remain related to Input A, but it is not necessarily methodologically identical to it.

Transformation Need

Every material transformation should have an identifiable reason.

A Transformation Need may arise because: the simulation requires another unit, sources use incompatible formats, sampling frequencies differ, missing information must be treated, data contains technical noise, model interfaces require normalization, multiple sources must be aligned, resolution must change, derived variables are required, or information must be adapted to the simulation representation.

Transformation Purpose

The transformation purpose should explain:

Why must this information be changed before simulation use?

Examples include:

Compatibility

Alignment

Noise Reduction

Feature Construction

Missing-Value Treatment

Resolution Adjustment

Unit Harmonization

Data Fusion

or:

Simulation Interface Preparation

Transformation Necessity

SDTM distinguishes:

Necessary Transformation

from:

Convenience Transformation.

A transformation may be technically convenient without being methodologically justified.

Where a transformation materially affects simulation behavior, convenience alone should not establish its legitimacy.

Transformation Method

A Transformation Method defines how the original information is converted into the transformed Simulation Input.

It may specify: operation, algorithm, formula, rule, threshold, parameter, model, software process, sequence, or another transformation mechanism.

Transformation Rule

A transformation should be sufficiently explicit that the relationship between:

Input Before Transformation

and:

Input After Transformation

can be understood and, where required, reconstructed.

Transformation Parameters

Material parameters should remain traceable.

For example:

a smoothing filter may require: window size, threshold, frequency, kernel, or another parameter.

Changing these parameters may materially change the resulting Simulation Input.

Data Cleaning

Cleaning may include: invalid-record removal, duplicate removal, formatting correction, impossible-value treatment, corrupted-record treatment, or consistency correction.

Cleaning should not become an invisible mechanism for removing inconvenient data. Cleaning $\neq$ Evidence Removal

SIMULOS® establishes:

Data should not be removed merely because it produces undesirable simulation behavior.

Where materially relevant records are excluded, the basis for exclusion should remain identifiable. Filtering

Filtering may remove: noise, frequencies, observations, events, values, or records.

SDTM requires material filtering logic to remain visible.

Filter Effect

A filter may improve usability while reducing information.

Therefore:

Cleaner Data $\neq$ More Complete Data

The effect of filtering should be assessed where material. Normalization

Normalization may transform values into a common range or representation.

For example:

0–100 $\rightarrow$ 0–1

The normalized value should remain connected to: original scale, transformation rule, and semantic meaning.

Standardization

Standardization may express values relative to a distribution.

The underlying basis should remain identifiable where the transformed values affect simulation interpretation.

Unit Conversion

Unit transformations must preserve the relationship:

Original Value + Original Unit $\rightarrow$ Conversion Rule $\rightarrow$ Converted Value + Target Unit

SDTM prevents converted values from becoming detached from their original

measurement basis.

Format Conversion

Data may move between: structured formats, numerical representations, image formats, coordinate systems, encoding systems, categorical structures, or simulation-specific schemas.

Technical conversion should preserve materially relevant meaning.

Semantic Preservation Principle

SIMULOS® establishes:

A transformation that preserves technical readability but changes material meaning is not semantically neutral. Aggregation

Aggregation combines multiple observations into a reduced representation.

Examples include:

second → minute

individual → group

sensor points → regional average

Aggregation may simplify simulation input while hiding: variation, extremes, sequences, correlations, or rare events.

Aggregation Loss

Where aggregation removes materially relevant information, the loss should remain visible.

For example:

an average temperature may hide short thermal spikes capable of affecting system failure. Disaggregation

Disaggregation converts aggregated information into a more granular representation.

Where the original information does not contain that granularity, disaggregated values may depend upon assumptions or models.

SIMULOS® establishes:

Disaggregation must not be represented as direct observation where the additional granularity was inferred. Interpolation

Interpolation estimates values between known observations.

Conceptually:

Observed A → Estimated Intermediate Value → Observed B

Interpolated values should remain distinguishable from direct observations where material. Extrapolation

Extrapolation estimates information beyond the observed range.

Because extrapolation extends beyond available observation, its methodological basis and limitations should remain visible.

No Extrapolation-as-Observation Principle

A value generated beyond the observed data range does not become observed reality merely because it is numerically precise. Resampling

Resampling changes the frequency, density, or structure of observations.

Examples include:

100 Hz → 10 Hz

or:

hourly → minute-level representation

Resampling may: remove information, introduce estimated information, alter signal behavior, or change temporal relationships.

Encoding

Information may be encoded into representations required by simulation systems.

For example:

Category → Numerical Vector

The encoded representation should remain traceable to the original semantic category.

Derived Input

A Derived Simulation Input is produced from one or more existing inputs.

Conceptually:

Input A + Input B + Rule → Derived Input C

The derivation relationship should remain preserved. Derived ≠ Observed

SIMULOS® establishes:

A calculated value must not silently acquire the status of a directly observed

value.

Feature Derivation

Simulation models may require derived features.

Examples include: velocity from position changes, acceleration from velocity, risk indicator from multiple variables, density from count and area, or another constructed feature.

The derivation method should remain traceable where material.

Missing-Value Treatment

Missing information may be treated through: exclusion, substitution, interpolation, imputation, default value, synthetic replacement, or another governed method.

SDTM requires the treatment to remain explicit. Missing $\neq$ Zero

SIMULOS® establishes:

Absence of information must not automatically become a zero-valued observation. Imputation

Imputation replaces missing information with an estimated value.

An imputed value should remain identifiable as:

Imputed

rather than:

Observed

or:

Measured.

Imputation Basis

The basis may include: statistical estimate, neighboring observations, historical values, model output, population average, or another method.

The selected basis may materially influence simulation behavior.

Default Values

A simulation may technically require a value where none is available.

A default value may be used where methodologically justified.

The default should remain identifiable as a default rather than being silently

treated as source data.

Synthetic Supplementation

Synthetic information may be introduced during transformation to supplement missing or weak data.

SDTM governs the transformation process that introduces it.

SIIM preserves its synthetic identity.

SDRM governs what reality it can legitimately represent.

Outlier Treatment

Outliers may be: retained, excluded, capped, transformed, separately modeled, or investigated.

Outlier treatment should depend upon the simulation purpose and data basis.

Outlier $\neq$ Error

SIMULOS® establishes:

An unusual observation is not automatically an invalid observation.

Rare values may represent precisely the conditions a simulation needs to examine.

Data Fusion

Multiple information sources may be combined into one Simulation Input.

Conceptually:

Source A + Source B + Fusion Method → Fused Input

The fusion process should preserve materially relevant source relationships and transformation logic. Fusion $\neq$ Independent Confirmation

Two sources combined into one input do not necessarily constitute independent evidence.

Their dependencies may remain relevant.

Spatial Transformation

Spatial information may require: coordinate conversion, map projection, spatial aggregation, interpolation, alignment, or resolution change.

Material spatial transformation should remain traceable where it can influence simulated relationships.

Coordinate Transformation

Coordinate-system changes must preserve the applicable reference system and transformation basis.

Errors in coordinate conversion can produce apparently valid but physically incorrect simulation inputs.

Dimensional Transformation

Information may be reduced or expanded across dimensions.

For example:

3D Reality → 2D Representation

or:

Multiple Variables → Composite Index

Such transformations may intentionally simplify reality.

The simplification should remain connected to the simulation's Reality Representation under RRS.

Transformation Sequence

Simulation data may undergo multiple transformations.

For example:

Raw Sensor Data

↓

Cleaning

↓

Filtering

↓

Unit Conversion

↓

Resampling

↓

Normalization

↓

Simulation Input

SDTM requires material sequence to remain reconstructable.

Transformation Order Matters

Transformations may not be commutative.

Conceptually:

Filter → Aggregate

may produce a different result from:

Aggregate → Filter

Where order materially affects the resulting input, transformation sequence becomes a governed property.

Transformation Dependency

One transformation may depend upon the output of another.

These dependencies should remain visible where they affect reproducibility or interpretation.

Transformation Chain

A Simulation Transformation Chain records the material progression:

Original Input → T1 → Intermediate Input → T2 → Intermediate Input → T3 →
Simulation-Usable Input Transformation Identity

Each material transformation may receive an identifiable reference.

For example:

SDT-0042

This allows simulation records to reference the exact transformation logic used.

Transformation Version

Transformation logic may change.

For example:

Filter v1

may become:

Filter v2

A simulation executed under v1 should remain distinguishable from one executed under v2.

No Silent Transformation Update Principle

SIMULOS® establishes:

Materially changing transformation logic must not silently rewrite the transformation basis of previous simulation executions.

Transformation Reproducibility

Where required by the simulation purpose, a transformation should be sufficiently documented to allow an authorized independent implementation to reproduce the transformed input from the same original basis.

Reproducibility may require: method, parameters, software version, transformation sequence, source input version, and relevant execution conditions.

Transformation Reversibility

Not every transformation is reversible.

For example:

aggregation may permanently discard detail.

SDTM therefore distinguishes:

Reproducibility

from:

Reversibility.

A transformation may be reproducible while information loss makes reversal impossible.

Information Loss

Transformation may remove information.

Loss may be:

Intentional

Incidental

Necessary

or:

Unrecognized

SDTM requires material information loss to remain visible.

Transformation Gain

Some transformations create additional information representations.

For example:

interpolation may create additional values.

But:

More values $\neq$ More observations.

Generated resolution should not be confused with additional empirical evidence.

Precision Inflation

Transformation may produce numerically precise outputs from imprecise inputs.

For example:

an estimated value may mathematically produce:

17.384729

This numerical precision does not establish equivalent informational certainty.

No Precision Inflation Principle

SIMULOS® establishes:

Transformation must not create an appearance of certainty or observational precision unsupported by the underlying information basis.

Transformation-Induced Bias

Transformation logic may systematically alter the information basis.

Examples include: selective filtering, biased imputation, asymmetric clipping, population-dependent normalization, or model-based reconstruction.

Material transformation-induced bias should be identified and carried forward.

Transformation-Induced Uncertainty

A transformation may introduce additional uncertainty.

For example: interpolation uncertainty, extrapolation uncertainty, imputation uncertainty, fusion uncertainty, conversion uncertainty, or model-derived uncertainty.

The subsequent SIUM — Simulation Input Uncertainty Module governs the consolidated input uncertainty.

SDTM identifies transformation-induced contributions to that uncertainty.

Transformation Effect Assessment

After transformation, the resulting input should be assessed for material changes in: meaning, range, distribution, resolution, completeness, representativeness, uncertainty, and simulation influence.

Representativeness Reassessment

Transformation may change representativeness.

For example:

aggressive filtering may remove rare conditions that SDRM previously considered represented.

Where transformation materially affects representation:

SDRM reassessment may be required.

Transformation Cannot Launder Weak Input

SIMULOS® establishes:

Transformation does not automatically cure an unrepresentative, uncertain, ambiguous, or otherwise limited input basis.

A weak source input may remain weak after sophisticated processing.

Transformation Limitation

A Transformation Limitation identifies a material restriction introduced or preserved by transformation.

Examples include: lost high-frequency behavior, inferred missing values, reduced spatial resolution, extrapolation beyond observed range, or uncertain source alignment.

Transformation Condition

A transformed input may be suitable only under explicit conditions.

For example:

Interpolated values suitable only where gaps do not exceed 5 seconds.

The condition should remain attached to downstream simulation use.

Transformation Status

A material transformation may receive a status such as:

Transformation Accepted

Conditionally Accepted

Transformation Restricted

Transformation Rejected

or:

Transformation Undetermined

The status applies to the transformation's suitability for the defined simulation use.

Transformed Input Status

The resulting Simulation Input may be classified as:

Simulation-Usable

Conditionally Simulation-Usable

Restricted

Not Simulation-Usable

or:

Suitability Undetermined

This status does not replace subsequent temporal or uncertainty governance.

Transformation Change

A material change may occur in: transformation method, parameters, software, source mapping, sequence, threshold, imputation rule, fusion logic, or another transformation property.

Material changes should trigger reassessment of affected transformed inputs.

Transformation Impact Propagation

Where transformation materially affects downstream simulation use, the effect should propagate.

Conceptually:

Transformation Change → Transformed Input Change → Scenario / Execution Effect
→ Output Effect → Reliance Effect

The later SIMULOS® Parent Standards govern those downstream consequences.

Point-in-Time Transformation Reconstruction

SDTM supports questions such as:

Which transformation was used for this execution?

Which parameters were active?

Which source input version entered it?

Which values were imputed?

Which observations were removed?

Which transformations introduced additional uncertainty?

Transformation Traceability

SDTM establishes:

Source Input

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Transformation Need

↓

Transformation Method

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Transformation Parameters

↓

Transformation Execution

↓

Intermediate States

↓

Transformed Input

↓

Transformation Effects



Conditions / Limitations



Simulation Use

Minimum Implementation Framework

1. Identify Material Transformations

Determine which input changes can materially affect simulation behavior or interpretation. 2. Establish Transformation Purpose

Record why each material transformation is required. 3. Define Transformation Method

Specify the applicable: operation, rule, algorithm, formula, parameters, or processing method.

4. Preserve Source Input Identity

Bind every material transformation to the exact input version from which it originates. 5. Execute and Record the Transformation

Preserve the material transformation sequence and resulting input state. 6. Identify Generated, Removed, or Modified Information

Determine whether the transformation: removes information, generates inferred information, changes resolution, changes distribution, changes meaning, or changes simulation influence.

7. Assess Transformation Effects

Evaluate material effects upon: representativeness, uncertainty, completeness, precision, semantics, and simulation applicability.

8. Govern Missing Values, Outliers, and Synthetic Supplementation

Ensure treatment remains explicit and traceable. 9. Assess Transformation Limitations

Identify information loss, extrapolation, imputation, bias, uncertainty, or other material restrictions. 10. Assign Transformation Status

Determine whether the transformation is: Accepted, Conditionally Accepted, Restricted, Rejected, or Undetermined.

11. Establish Transformed Input Status

Determine whether the resulting input may proceed to subsequent SIDS governance. 12. Preserve Transformation Traceability

Maintain the complete relationship:

Source Input → Transformation → Transformed Input → Transformation Effect →
Simulation Use

Governance Outputs

SDTM may produce: Simulation Transformation Requirement, Transformation Purpose Record, Transformation Identifier, Transformation Method Record, Transformation Rule, Transformation Parameter Record, Data Cleaning Record, Filtering Record, Normalization Record, Standardization Record, Unit Conversion Record, Format Conversion Record, Aggregation Record, Disaggregation Record, Interpolation Record, Extrapolation Record, Resampling Record, Encoding Record, Derived Input Record, Feature Derivation Record, Missing-Value Treatment Record, Imputation Record, Default-Value Record, Synthetic Supplementation Record, Outlier Treatment Record, Data Fusion Record, Spatial Transformation Record, Coordinate Transformation Record, Dimensional Transformation Record, Transformation Sequence Record, Simulation Transformation Chain, Transformation Dependency Map, Transformation Version Record, Transformation Reproducibility Record, Information Loss Record, Transformation-Induced Bias Record, Transformation-Induced Uncertainty Record, Transformation Effect Assessment, Representativeness Reassessment Signal, Transformation Limitation Record, Transformation Condition Record, Transformation Status, Transformed Input Status, Transformation Change Record, and Simulation Data Transformation Traceability Map.

Use Case 1 — Autonomous Vehicle Sensor Simulation

Scenario

A simulation consumes vehicle sensor data sampled at different frequencies: camera — 30 Hz, radar — 20 Hz, lidar — 10 Hz, vehicle telemetry — 100 Hz.

The simulation requires a synchronized input stream. Application

SDTM governs:

Source Inputs

↓

Timestamp Alignment

↓

Resampling

↓

Interpolation Where Required

↓

Synchronized Simulation Inputs

Interpolated values remain distinguishable from direct measurements, and the transformation parameters remain traceable. Result

The simulation can determine exactly how the synchronized input stream was produced and which portions contain transformed rather than directly observed information. Use Case 2 — Industrial Failure Simulation

Scenario

Historical sensor records contain missing values and extreme temperature spikes.

A preprocessing process proposes to: replace missing values with averages, remove extreme values as outliers, smooth the resulting signal.

Application

SDTM prevents automatic treatment.

The temperature spikes are assessed before removal because they may represent precisely the failure conditions the simulation intends to examine.

Missing values remain distinguishable from observed measurements after imputation. Result

The transformation produces a governed Simulation Input without silently removing potentially material failure behavior or representing imputed values as direct observations.

Architectural Position

Simulation Data Transformation is the third internal governance space of the Simulation Input & Data Standard (SIDS).

The complete SIDS progression is:

Simulation Input Identification

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Simulation Data Representativeness

↓

Simulation Data Transformation

↓

Simulation Temporal Data



Simulation Input Uncertainty

SIIM establishes:

What is the input?

SDRM establishes:

What can it legitimately represent?

SDTM establishes:

What happened to it before simulation use?

The next module, STDM — Simulation Temporal Data Module, establishes whether the information and its transformed state are temporally appropriate for the simulation context in which they will be used.

Foundational Principle

A transformed Simulation Input remains legitimate only when the path from its original information basis to its simulation-usable form remains sufficiently identifiable, justified, reproducible, limitation-aware, and traceable, without allowing transformation to manufacture unsupported meaning, representativeness, observational status, precision, or certainty.

Canonical Closing Statement

Simulation Data Transformation Module establishes the transformation governance layer of the Simulation Input & Data Standard by governing how identified Simulation Inputs are cleaned, filtered, normalized, converted, aggregated, interpolated, extrapolated, resampled, encoded, derived, imputed, fused, supplemented, aligned, or otherwise transformed for simulation use. Through explicit transformation purpose, method, parameters, sequencing, source binding, versioning, information-loss assessment, generated-information distinction, bias and uncertainty identification, representativeness reassessment, conditions, limitations, status, and complete transformation traceability, SDTM ensures that simulation-ready information remains methodologically connected to the information from which it originated and that preprocessing cannot silently manufacture properties the underlying data never possessed.

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